



Modelling Financial Markets

Chapter 16: Digital Assets and DeFi



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Acronyms used in this material (1/2)

- AIC: Akaike Information Criterion
- AMM: Automated Market Maker
- AP: Authorised Participant
- AR: AutoRegressive (model)
- ATM: Automated Teller Machine
- BET: Bucharest Exchange Trading (index)
- BIC: Bayesian Information Criterion
- BNB: BNB (native token of BNB Chain, formerly Binance Coin)
- BSC: BNB Smart Chain
- COVID-19: COronaVirus Disease 2019
- CRIX: CRypt Index
- DAI: Dai (stablecoin of the Maker/Sky protocol)
- ES: Expected Shortfall
- ETF: Exchange-Traded Fund
- ETHA: iShares Ethereum Trust ETF (ticker)
- FDIC: Federal Deposit Insurance Corporation
- LP: Liquidity Provider
- LUNA: Luna (native token of the Terra blockchain)
- LVR: Loss-Versus-Rebalancing
- MDD: Maximum DrawDown
- NAV: Net Asset Value
- PC1: first Principal Component
- PC2: second Principal Component
- PCA: Principal Component Analysis
- SE: Standard Error
- SVB: Silicon Valley Bank

Acronyms used in this material (2/2)

- TE: Tracking Error
- TVL: Total Value Locked
- US: United States
- USDC: USD Coin (stablecoin issued by Circle)
- UST: TerraUSD (algorithmic stablecoin of the Terra blockchain)
- XRP: XRP (the native token of the XRP Ledger)

Outline

Motivation

Stylised Facts of Crypto Returns

Crypto Indices: CRIX

Stablecoins

DeFi and Automated Market Makers

Spot ETFs and Crypto Equities

Are Cryptos Becoming Alternative Assets?

Wrap-up

References

Why Digital Assets in a Course on Market Modelling?

The question of this chapter

- Crypto-assets trade 24/7, have their own indices, stablecoins and exchanges without intermediaries
- Since January 2024 they also trade on US exchanges through spot exchange-traded funds
- Are they a separate asset class, a risk asset like technology shares, or a hedge?

What we do in this chapter

- Measure the market and compare its stylised facts with equities and gold
- Build a crypto index (CRIX) and choose its number of constituents
- Model stablecoins, automated market makers and spot ETFs with real data
- Test whether cryptos are becoming alternative assets

Bitcoin: Money Without a Central Issuer



Bust of Satoshi Nakamoto, Graphisoft Park,
Budapest

Photo: Fekist (2021); CC BY-SA 4.0; Wikimedia
[Commons](#)

- Nakamoto (2008): a peer-to-peer electronic cash system under the pseudonym Satoshi Nakamoto
 - ▶ Transactions are grouped in **blocks**; each block contains the hash of the previous one: a **blockchain**
 - ▶ **Proof of work**: miners compete to find a valid block; the longest chain is the valid history
- New issuance halves over time until a total of 21 million coins
[Bitcoin.org FAQ](#); 20.09 million existed on 18 Sep 2026
- Technical survey: [Lin, Pele & Ren \(2025\)](#); economics: [Härdle, Harvey & Reule \(2020\)](#)

Mining: the Cost of Securing the Ledger



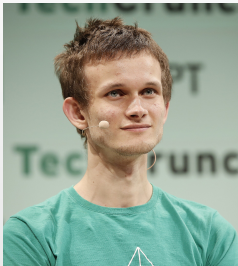
A Bitcoin mining farm (2014)

Photo: Marko Ahtisaari (2014); CC BY 2.0; Wikimedia

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- Each block pays the miner a **block reward** plus transaction fees
 - ▶ The reward is halved over time: new supply shrinks by design
[Bitcoin.org FAQ](#)
- Specialised hardware, cheap electricity: mining is an industry with its own cycles
- Ethereum replaced proof of work with **proof of stake** on 15 September 2022 [ethereum.org](#): [The Merge](#): validators lock coins as collateral
 - ▶ Staked coins earn a yield: a first link between crypto and interest rates

Ethereum: a Programmable Blockchain



Vitalik Buterin, co-founder of Ethereum, London
2015

Photo: John Phillips (2015); CC BY 2.0; Wikimedia
[Commons](#)

- Buterin (2014): a blockchain that runs programs, called **smart contracts**
 - ▶ A smart contract holds assets and moves them by fixed rules, without an intermediary
- **Tokens**: assets issued by smart contracts on an existing blockchain (stablecoins, governance tokens, tokenised funds)
- **Decentralised finance (DeFi)**: exchanges, lending and derivatives built from smart contracts [Schär \(2021\)](#)
- Ether is also the collateral and the “fuel” (gas) that pays for computation

Key Terms of This Chapter

- ▣ **Crypto-asset:** a digital asset recorded on a blockchain; **coin** = native asset of a blockchain (Bitcoin, Ether), **token** = asset issued by a smart contract
- ▣ **Stablecoin:** a token designed to keep a fixed price, usually 1 USD Lyons & Viswanath-Natraj (2023)
- ▣ **Automated market maker (AMM):** a smart contract that quotes prices from a formula applied to its reserves Adams, Zinsmeister & Robinson (2020)
- ▣ **Total value locked (TVL):** the market value of the assets deposited in DeFi smart contracts DefiLlama
- ▣ **Spot ETF:** an exchange-traded fund that holds the crypto-asset itself, not futures
- ▣ **Regulation:** MiCA, Regulation (EU) 2023/1114 in the European Union, GENIUS Act, Public Law 119-27 for payment stablecoins in the US

Data Used in This Chapter

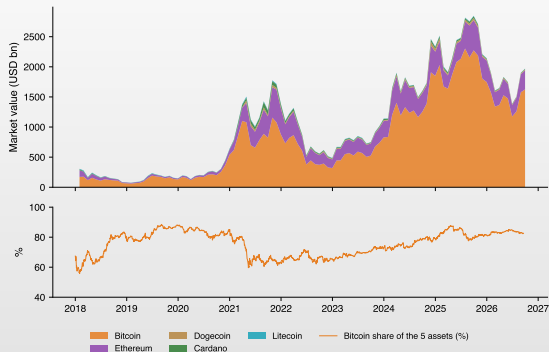
Series	Content	Period
Crypto prices	Bitcoin, Ethereum, Solana, Dogecoin, Cardano, Litecoin; daily, 7 days a week	2018–2026
Stablecoins	USDT, USDC, DAI prices; supply by coin (DeFiLlama)	2019–2026
Coin supply	Coins in circulation, daily (Coin Metrics Community Data)	2018–2026
DeFi	Total value locked, all chains and by chain (DeFiLlama)	2020–2026
ETFs and shares	spot ETFs on Bitcoin (IBIT) and Ether (ETHA), Coinbase, Strategy; S&P 500, Nasdaq 100, gold	2018–2026
Asset universe	26 assets: cryptos, equity indices, currencies, commodities, bonds	2019–2026

- Daily closes up to 18 September 2026; crypto on its own 7-day calendar; joint analyses on common days (prices joined first, then returns)

Market Value: Definition and a Worked Example

- ▣ **Market value** (market capitalisation) of coin i on day t : $M_{i,t} = P_{i,t} \times Q_{i,t}$
 - ▶ $P_{i,t}$: price in USD; $Q_{i,t}$: coins in circulation (supply)
- ▣ Bitcoin on 18 Sep 2026: $P = 80\,901$ USD, $Q = 20.09$ million
 - ▶ $M = 80\,901 \times 20.09 \times 10^6 \approx 1625$ billion USD, i.e. 1.63 trillion
- ▣ Ether: $P = 2\,611$ USD, $Q = 122.1$ million, $M \approx 319$ billion USD
- ▣ **Free float** matters: supply held by the issuer or locked is not traded
 - ▶ XRP and Chainlink report a supply that includes tokens held by the issuer; we leave them out of the index universe

Five Coins, One Dominant Asset



- Market value of Bitcoin, Ethereum, Dogecoin, Cardano and Litecoin (month ends) and Bitcoin's share of the five

Reading the Market-Value Chart

- The five coins were worth 1970 billion USD on 18 Sep 2026; peak 3135 billion on 6 Oct 2025
 - ▶ The market value moves with prices, not with supply: supply changes slowly
- Bitcoin's share: 82.5% today, between 55.8% (1 Feb 2018) and 88.5% (5 Sep 2019)
- **Herfindahl–Hirschman index** of concentration: $\text{HHI} = \sum_i w_i^2$, with w_i the market-value shares
 - ▶ Today $\text{HHI} = 0.71$: the “effective number” of coins $1/\text{HHI}$ is only 1.42
- A crypto portfolio weighted by market value is mostly a Bitcoin position

Recap: What Digital Assets Are

- ▣ A blockchain is a shared ledger secured by proof of work or proof of stake
- ▣ Coins are native assets; tokens, stablecoins and DeFi protocols are built by smart contracts
- ▣ Market value = price $\times$ supply in circulation; free float matters
- ▣ The market is highly concentrated: Bitcoin holds about four fifths of our five-coin universe

Indicators Used in This Section (1/2)

- Daily log return in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$, each series on its own calendar
 - ▶ m = observations per year: about 365 for crypto (7 days a week), about 252 for shares and indices
- **Annualised volatility**: $\sigma_a = s\sqrt{m}$, with s the standard deviation of daily returns
- **Skewness** $S = \overline{(r - \bar{r})^3} / s^3$ and **excess kurtosis** $K = \overline{(r - \bar{r})^4} / s^4 - 3$ ($K = 0$ for the Normal distribution)
- **Maximum drawdown**: $MDD = \min_t (P_t / \max_{u \leq t} P_u - 1)$, the worst fall from a previous peak
- **Volatility clustering**: first-order autocorrelation of $|r_t|$, $\hat{\rho}_1(|r|)$ (Chapter 1)

Indicators Used in This Section (2/2)

- **Tail index** (Hill estimator) from the k largest losses $L_{(1)} \geq \dots \geq L_{(k+1)}$
 - ▶ $\hat{\alpha}_H = \left(\frac{1}{k} \sum_{j=1}^k \ln \frac{L_{(j)}}{L_{(k+1)}} \right)^{-1}$, with $k = 5\%$ of the losses; smaller $\hat{\alpha}_H =$ heavier tail
- **VaR 1% and ES 2.5%** (Chapter 7), historical, daily, in % of the position
 - ▶ $\text{VaR}_{1\%} = -q_{1\%}(r)$; $\text{ES}_{2.5\%} = -\overline{r \mid r \leq q_{2.5\%}(r)}$
- All indicators are computed on 2018–2026 (Solana from April 2020, when its price history starts)
- Comparison assets: S&P 500 (US shares) and gold (the classic alternative asset)

Crypto vs Shares and Gold, 2018–2026

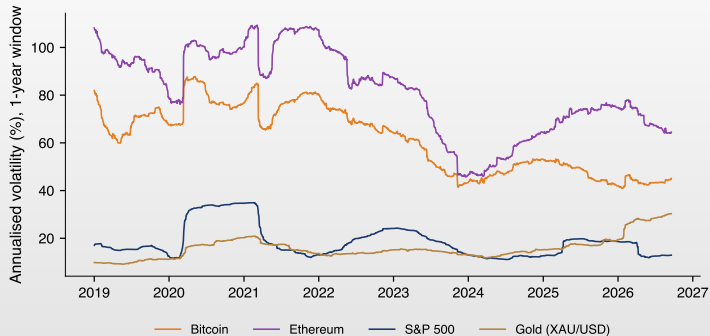
	n	σ_a (%)	S	K	MDD (%)	$\hat{\rho}_1(r)$	$\hat{\alpha}_H$	VaR _{1%}	ES _{2.5%}
Bitcoin	3 182	64	−0.95	14.8	−82	0.17	2.73	10.2	10.3
Ethereum	3 182	84	−0.82	11.1	−94	0.16	2.67	13.5	13.7
Solana	2 351	117	−0.22	7.9	−96	0.28	2.85	14.5	17.1
Dogecoin	3 182	124	4.88	106.4	−92	0.37	2.83	13.8	16.4
S&P 500	2 189	19	−0.65	14.9	−34	0.36	2.87	3.4	3.8
Gold	2 269	17	−0.55	10.3	−28	0.13	2.75	3.0	3.2

□ Daily log returns in %; VaR and ES as daily losses in % of the position; $\hat{\alpha}_H$ from the largest 5% of losses

Reading the Table

- **Volatility:** Bitcoin 64% a year, 3.3 times the S&P 500; Solana and Dogecoin above 100%
 - ▶ Daily VaR 1% of Bitcoin: 10.2%, 3.0 times that of the S&P 500
- **Drawdowns:** Bitcoin lost 82% from its peak, Ethereum 94%, Solana 96%; the S&P 500 at most 34%
- **Heavy tails everywhere:** Hill indices between 2.6 and 2.9 for all six assets: crypto is more volatile, not necessarily heavier-tailed once scaled
- **Dogecoin:** skewness 4.88 and kurtosis 106.4: a few social-media-driven jumps dominate the moments
- **Worst Bitcoin day:** -46.5% on 12 Mar 2020 (COVID-19 crash), four days before the worst S&P 500 day (16 Mar 2020, -12.8%)

Volatility Over Time

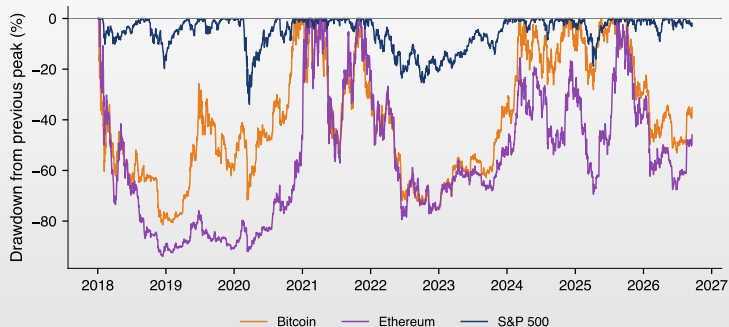


□ Annualised standard deviation of daily log returns over a moving one-year window

Reading the Volatility Chart

- ▣ Bitcoin's one-year volatility fell from 88% to 45% today; Ethereum from 109% to 64%
 - ▶ A maturing market: deeper liquidity, institutional holders, ETFs
- ▣ Bitcoin is still 3.5 times as volatile as the S&P 500 (13%)
- ▣ Gold's one-year volatility reached 30% in 2026, a record for the sample: the “safe” asset is not always calm
- ▣ Ethereum is persistently more volatile than Bitcoin, with a market value about five times smaller

Drawdowns: How Deep and How Long

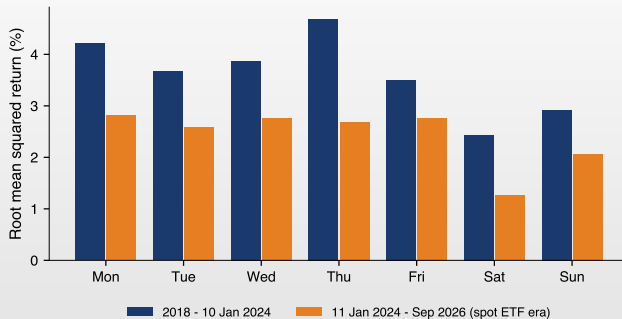


□ Fall from the previous maximum price, in %

Reading the Drawdown Chart

- ▣ Bitcoin was more than 50% below its previous peak on 36% of the days since 2018; Ethereum on 61%
 - ▶ The S&P 500 never fell 50% in this period; its worst drawdown was 34% in March 2020
- ▣ Today: Bitcoin 35%, Ethereum 46% below their peaks
- ▣ Long “crypto winters” (2018–2020, 2022–2023) follow each boom: a reason to study bubbles (Chapter 17)
- ▣ For an investor, drawdown duration matters as much as volatility: capital is tied up for years

A 7-Day Market with a Weekend Lull



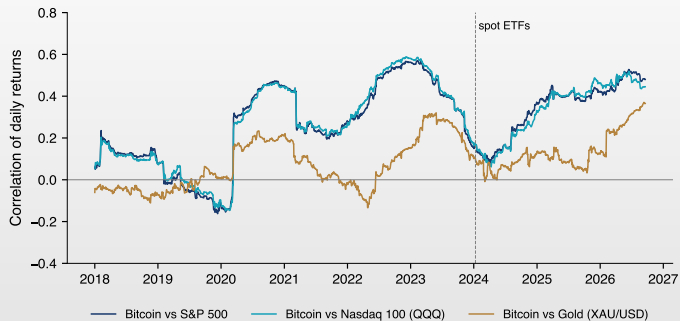
- Root mean squared daily Bitcoin return by day of the week, before and after the spot ETFs started trading

 MFM_ch16_weekday

Reading the Day-of-Week Chart

- Weekend variance divided by weekday variance: 0.45 before and 0.39 after 11 January 2024
 - ▶ Bitcoin trades on Saturdays, but the information flow (macro news, US equity trading, ETF flows) does not
- All days became calmer after the ETF launch: weekday root mean square from 4.01% to 2.73%
- Consequence for modelling: annualise crypto volatility with 365 days, but do not treat all days as identical
- Is the weekend ratio different after the ETFs? A block-bootstrap test in Seminar 16 (B1)

Is Bitcoin Moving with Shares?



- Correlation of daily log returns over 250 common trading days (prices joined on common days first)

Reading the Correlation Chart

- Bitcoin vs S&P 500: 0.02 in 2017–2019, 0.39 in 2020–2023, 0.38 since the ETFs (674 days)
 - ▶ The break came in March 2020, not in January 2024: Bitcoin became a risk asset during the pandemic
- Nasdaq 100 behaves like the S&P 500 (0.38 since 2024): no special link with technology shares
- Gold: -0.02 , 0.12 , 0.22 : rising, 0.37 in the last window
- Diversification benefit of Bitcoin against shares is smaller than before 2020 [Bouri et al. \(2017\)](#), [Liu & Tsyvinski \(2021\)](#)

Recap: Stylised Facts

- ▣ Crypto is 3–7 times as volatile as shares, with drawdowns of 80–96%
- ▣ Heavy tails and volatility clustering, as in every financial series (Chapter 1)
- ▣ Volatility falls as the market matures; weekends are calmer than weekdays
- ▣ Since 2020 Bitcoin is positively correlated with shares, less so with gold

Why a Crypto Index?



Main building of Humboldt-Universität zu Berlin, where
CRIX was developed

Photo: [Diego Delso \(2016\)](#); CC BY-SA 4.0; Wikimedia

[Commons](#)

- A benchmark for fund managers and a basis for investable products
- **CRIX** [Trimborn & Härdle \(2018\)](#): a market-value weighted index of the crypto market
 - ▶ The number of constituents is **not fixed**: it is chosen by the Akaike information criterion (AIC) so that the index tracks the whole market
 - ▶ Reason: coins appear and vanish daily; a fixed number (like 30 or 500 for shares) would be arbitrary
- Published at thecrix.de; a review of crypto indices: [Găman, Pele, Härdle & Będowska-Sójka \(2026\)](#)

A Market-Value Weighted (Laspeyres) Index

- Level on day t in reallocation period τ : $I_t = \frac{\sum_{i=1}^k P_{i,t} Q_{i,\tau}}{D_\tau}$
 - ▶ $Q_{i,\tau}$: coins in circulation, fixed during the period; D_τ : the **divisor**
- Equivalent form: the index return is the weighted average of the constituents' returns
 - ▶ $\frac{I_t}{I_{t_0}} = \sum_i w_i \frac{P_{i,t}}{P_{i,t_0}}$, $w_i = \frac{P_{i,t_0} Q_{i,\tau}}{\sum_j P_{j,t_0} Q_{j,\tau}}$ (market-value shares at the start of the period)
- At a **reallocation** (new Q , new members) the divisor changes so that the level does not jump
 - ▶ $D_{\tau+1} = \frac{\sum_i P_{i,t} Q_{i,\tau+1}}{I_t}$: old and new baskets give the same I_t on the switch day
- In this chapter: monthly reallocation, weights from the market values at the previous month end

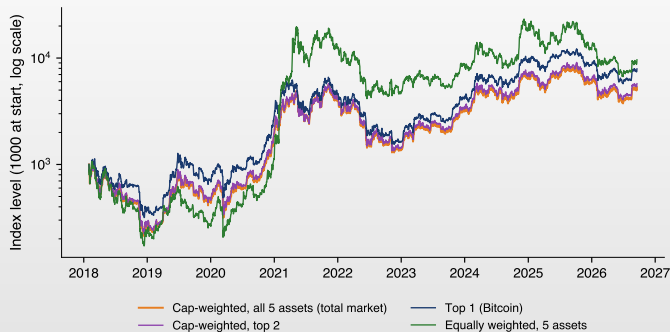
Worked Example: a Two-Coin Index

- Coin A: $P_0 = 50\,000$, $Q = 19.8$ million, $M_A = 990$ billion; coin B: $P_0 = 2\,500$, $Q = 120$ million, $M_B = 300$ billion
 - ▶ Start at 1000: $D_0 = (M_A + M_B)/1000 = 1\,290$ million; weights $w_A = 76.7\%$, $w_B = 23.3\%$
- A month later: A rises 10% to 55 000, B falls 10% to 2 250
 - ▶ $I_1 = (55\,000 \cdot 19.8 + 2\,250 \cdot 120) \cdot 10^6 / D_0 = 1053.5$
 - ▶ Check with weights: $76.7\% \cdot 10\% - 23.3\% \cdot 10\% = 5.35\%$, the same as $I_1/1000 - 1$
- An equally weighted index would show 0%: weighting choice matters as much as constituent choice
- Replacing a constituent or updating Q : see Seminar 16, A1 and A2

How Many Constituents? The AIC Rule

- Total-market index I^{tot} (all coins) and index $I^{(k)}$ with the k largest coins
 - ▶ Tracking difference: $e_t^{(k)} = \Delta \ln I_t^{(k)} - \Delta \ln I_t^{\text{tot}}$, $\hat{\sigma}_k^2 = \frac{1}{n} \sum_t (e_t^{(k)})^2$
- Choose k that minimises $AIC(k) = n \ln \hat{\sigma}_k^2 + 2k$, in the spirit of [Trimborn & Härdle \(2018\)](#)
 - ▶ Each extra coin must reduce $n \ln \hat{\sigma}_k^2$ by more than 2 to be admitted
 - ▶ Bayesian information criterion (BIC): penalty $k \ln n$ instead of $2k$, hence fewer coins
- Tracking error per year: $TE_k = \hat{\sigma}_k \sqrt{365}$, in %
- Our universe: Bitcoin, Ethereum, Dogecoin, Cardano, Litecoin, the coins with a public daily supply equal to the circulating supply

Crypto Indices Since 2018

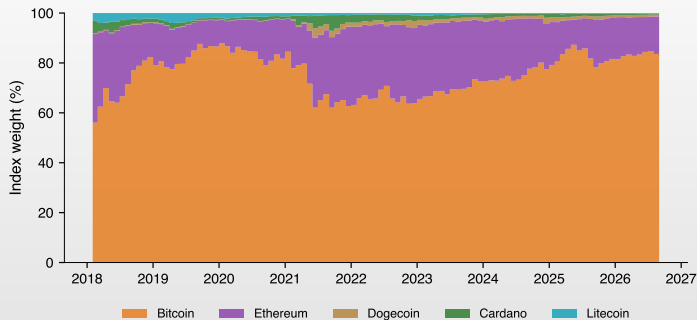


- Monthly reallocation; weights from market values at the previous month end; log scale

Reading the Index Chart

- Total-market index: 21.4% a year over 8.6 years, volatility 65.7%
 - ▶ The compound annual growth rate is $(I_T/I_0)^{1/\text{years}} - 1$
- Bitcoin alone did better than the market-value index: the other coins lost weight and value
- Equal weights: 30.0% a year, but volatility 82.5%
 - ▶ Driven by one year: 2021 (+998% vs +113%), when Dogecoin and Cardano soared; a few extreme months, not a stable premium
- An index is only as good as its universe: survivors today are not the leaders of 2018 [Będowska-Sójka, Wójcik & Pele \(2026\)](#)

Index Weights Over Time

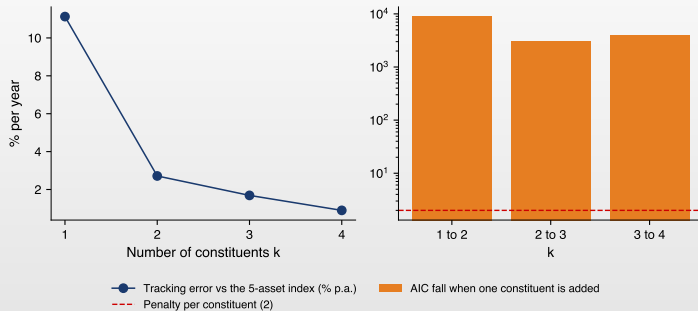


□ Market-value weights at each monthly reallocation, in %

Reading the Weights

- ▣ Bitcoin's weight: 75.7% on average, never below 56.1%; today 82.9%
- ▣ Ethereum 15.8%; Dogecoin, Cardano and Litecoin together below 2%
- ▣ The 2021 bull market briefly lifted the smaller coins, then Bitcoin regained its share
 - ▶ Altcoins (coins other than Bitcoin) gain weight late in booms and lose it in busts
- ▣ Concentration makes crypto indices behave like Bitcoin with a small add-on

Tracking Error and the AIC Rule



□ Left: tracking error of the top- k index against the five-coin index; right: fall in AIC when the next coin is added, log scale

Reading the AIC Chart

- Bitcoin alone tracks the five-coin market with an error of 11.1% a year; the top two with 2.7%
 - ▶ Top three: 1.7%; top four: 0.9%
- Adding coin 2 lowers AIC by 8 893, coin 3 by 2 991, coin 4 by 3 979: thousands of times the penalty of 2
- With only five candidates, AIC keeps 4 of 4 coins in every year: the rule bites only when the market has a long tail of small coins
 - ▶ CRIX applies it to a much larger universe, where the smallest coins barely move the total [Trimborn & Härdle \(2018\)](#)
- Project idea: extend the universe with more coins and redo the selection

Recap: Crypto Indices

- ▣ A Laspeyres index with a divisor: fixed quantities within a period, continuity at reallocation
- ▣ CRIX chooses its number of constituents by AIC, trading tracking accuracy against size
- ▣ Market-value weighting makes crypto indices mostly Bitcoin; equal weights add risk, not a stable premium
- ▣ Free float and survivorship are the two practical traps

Three Ways to Keep a Token at 1 USD

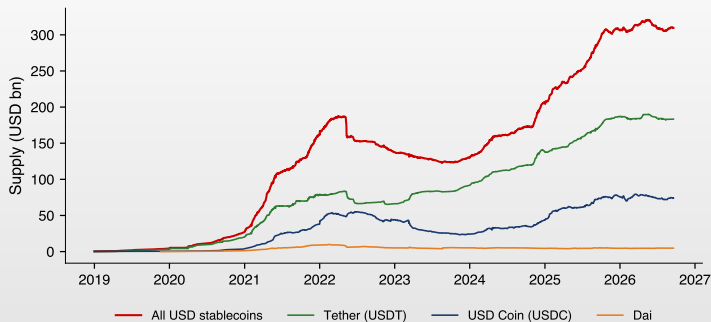
Design	Backing	Examples (supply, USD bn)
Fiat-backed	Bank deposits and short US Treasury bills held by the issuer	USDT 183.48, USDC 75.48, PYUSD 2.79
Crypto-backed	Over-collateralised crypto deposits or hedged positions, managed by smart contracts	USDS 6.51, USDe 4.90, DAI 4.78
Algorithmic	No external collateral: supply expands and contracts against a second token	TerraUSD (collapsed in May 2022)

- All USD stablecoins: 310 billion USD; fiat-backed 286 billion, crypto-backed 26.3 billion, algorithmic 0.66 billion ([DefiLlama](#), September 2026)
- Regulation: reserves, redemption and disclosure rules in [MiCA](#), [Regulation \(EU\) 2023/1114](#) and [GENIUS Act](#), Public Law 119-27

Keeping the Peg: Arbitrage Bands

- Authorised traders can **mint** (pay 1 USD, receive 1 coin) and **redeem** (return 1 coin, receive 1 USD) with the issuer
 - ▶ With a redemption cost c_r and a minting cost c_m (fees, settlement time), arbitrage keeps the market price P in the band $1 - c_r \leq P \leq 1 + c_m$
- Below the band: buy on the market, redeem at 1 USD; above: mint at 1 USD, sell on the market
- The band breaks when redemption stops (bank holiday, frozen reserves) or when the backing is doubted
 - ▶ Evidence on arbitrage and issuance: [Lyons & Viswanath-Natraj \(2023\)](#); on the link between Tether issuance and Bitcoin prices: [Griffin & Shams \(2020\)](#)
- A worked arbitrage calculation: Seminar 16, A7

The Rise of Stablecoins



Supply in circulation, billion USD (DefiLlama)

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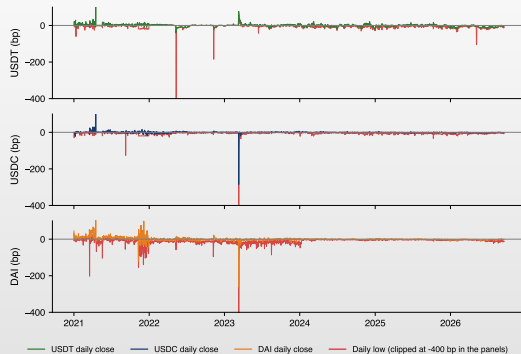
Reading the Supply Chart

- ▣ From 4.2 billion USD in January 2020 to 310 billion today: about 74 times
 - ▶ Stablecoins are the cash leg of crypto trading and DeFi, and a channel for dollar payments
- ▣ Tether: 183 billion (59% of the total); USD Coin: 74 billion (24%); Dai: 4.8 billion
- ▣ The 2022–2023 dip: the total fell from 187 billion (2 Apr 2022) to 123 billion (19 Aug 2023); TerraUSD vanished and USD Coin fell from 55 to 23 billion
- ▣ Reserves of payment stablecoins sit in cash-like assets such as short-term Treasury bills ([GENIUS Act, Public Law 119-27](#)): stablecoins now matter for the money market

Measuring the Peg

- Deviation in basis points (1 bp = 0.01%): $d_t = 10^4 (P_t - 1)$, from the daily close; the daily low shows intraday stress
 - ▶ Share of days with $|d_t| > 50$ bp; largest fall of the daily low
- Speed of return to the peg: AR(1) model $d_t = c + \phi d_{t-1} + \varepsilon_t$
 - ▶ **Half-life** $h = \ln 0.5 / \ln \phi$ (days): after h days half of a deviation is gone
 - ▶ Example: $\phi = 0.5$ gives $h = 1$ day; $\phi = 0.9$ gives $h = 6.6$ days
- Sample 2021–2026: earlier quotes are thin and noisy

Deviations from 1 USD



- Daily close and daily low minus 1 USD, in basis points, 2021–2026; lows below -400 bp are cut at the panel edge

Reading the Peg Chart

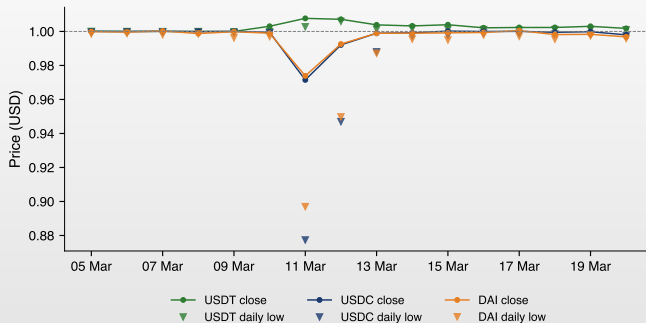
	s.d. (bp)	$ d_t > 50$ (%)	lowest close (bp)	lowest low (bp)	h (days)
Tether	7.3	0.14	-41	-515	2.2
USD Coin	7.5	0.14	-285	-1226	0.6
Dai	10.8	0.57	-261	-1030	0.6

- In normal times the peg holds within a few basis points; deviations halve within 0.6 to 2.2 days
- Stress is rare but deep: USD Coin's daily low reached -1226 bp on 11 Mar 2023; Tether's -515 bp on 12 May 2022
- Holding a stablecoin is like holding a claim with a short put on the reserves: nothing happens most days, a large loss is possible on a few

March 2023: a Bank Failure Hits a Stablecoin

- Friday 10 March 2023: Silicon Valley Bank (SVB) is closed and the Federal Deposit Insurance Corporation (FDIC) is appointed receiver [FDIC \(2023\)](#)
 - ▶ USD Coin relied on reserves held at SVB, so holders feared a loss on the backing [Diop, Chevallier & Sanhaji \(2024\)](#)
- Weekend: banks closed, redemptions could not settle, so the arbitrage band disappeared
- Sunday 12 March: the Treasury, the Federal Reserve and the FDIC announce that all SVB depositors are protected, with access from Monday [Treasury, Federal Reserve & FDIC \(2023\)](#)
- The peg returned within two days: the event tests the backing, not the design

USD Coin and Dai Below Par, Tether Above



□ Daily close (lines) and daily low (triangles), 5–20 March 2023

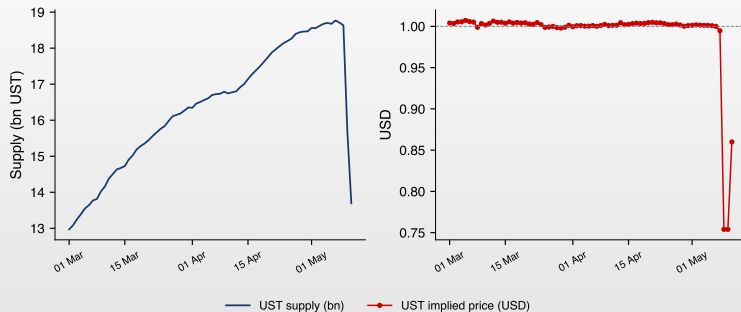
Reading the March 2023 Chart

- ▣ USD Coin: daily low 0.8774 USD on Saturday 11 March, close 0.9715
 - ▶ Dai fell with it (low 0.8970): a crypto-backed coin hit through its collateral [Diop, Chevallier & Sanhaji \(2024\)](#)
- ▣ Tether rose to 1.0077 at the close (high 1.0296): a flight to the other large stablecoin
- ▣ Contagion from traditional banking to crypto, the reverse of the usual worry
- ▣ For risk models: peg deviations have extreme kurtosis; a VaR from normal days is meaningless here

TerraUSD: an Algorithmic Stablecoin Collapses

- TerraUSD (UST) kept its peg by exchange with the Terra token LUNA, not with reserves
 - ▶ Demand came from a lending protocol (Anchor) that promised high yields to UST depositors [Liu, Makarov & Schoar \(2023\)](#)
- Terra, the third largest crypto ecosystem after Bitcoin and Ethereum, collapsed in three days in May 2022 and wiped out 50 billion USD in valuation [Liu, Makarov & Schoar \(2023\)](#)
- After a few large holders moved on 7 May, others followed: a bank run, visible on the blockchain in real time
- Wealthier, more sophisticated investors ran first and lost less [Liu, Makarov & Schoar \(2023\)](#)

The Run on TerraUSD



Supply and implied price (market value / supply), March–May 2022 (DefiLlama); the series ends on 11 May 2022

Reading the TerraUSD Chart

- ▣ Supply grew steadily to 18.8 billion on 7 May 2022; four days later it was 13.7 billion: 27% of the coins were redeemed
- ▣ Implied price fell to 0.754 USD on 10 May 2022; redemption into LUNA created more LUNA and pushed its price down: a death spiral
- ▣ Without external collateral, the peg rests on confidence alone: a run is self-fulfilling
- ▣ Lesson for regulators: payment stablecoins must hold liquid reserves ([GENIUS Act](#), [Public Law 119-27](#), [MiCA](#), [Regulation \(EU\) 2023/1114](#))

Tokenisation of Real-World Assets

- ▣ **Tokenised funds:** shares of a money-market or Treasury fund issued as tokens on a blockchain
 - ▶ Examples (supply, billion USD, September 2026, [DefiLlama](#)): Circle USYC 2.51, BlackRock BUIDL 2.29, Ondo USDY 2.21
- ▣ Unlike a payment stablecoin, they pay the Treasury yield to the holder
- ▣ Benefits: settlement 24/7, use as collateral in DeFi; risks: smart-contract bugs, legal ownership, liquidity in stress [Aramonte, Huang & Schrimpf \(2021\)](#)
- ▣ Token economics and adoption: [Cong, Li & Wang \(2021\)](#)

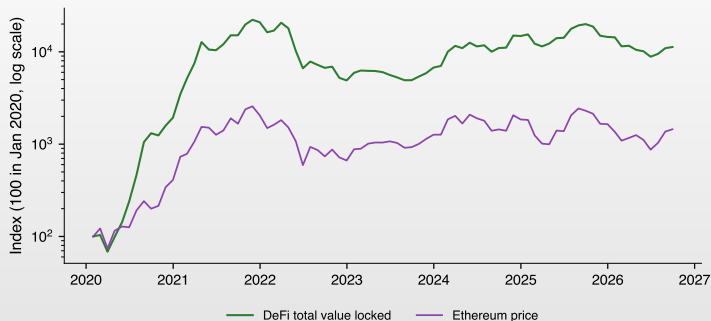
Recap: Stablecoins

- ▣ Three designs: fiat-backed, crypto-backed, algorithmic; fiat-backed dominate
- ▣ The peg is an arbitrage band that holds only while redemption works
- ▣ Normal days: deviations of a few basis points with a half-life below three days; stress days: hundreds of basis points
- ▣ Two runs: USD Coin in March 2023 (a reserve shock, recovered), TerraUSD in May 2022 (no reserves, collapsed)

Decentralised Finance in Five Layers

- Schär (2021) describes DeFi as a stack of layers
 - ▶ **Settlement:** the blockchain and its native coin (Ether)
 - ▶ **Assets:** tokens issued on it (stablecoins, governance tokens)
 - ▶ **Protocols:** exchanges, lending, derivatives as smart contracts
 - ▶ **Applications and aggregators:** user interfaces that route trades across protocols
- Promise: open access, transparency, composability (protocols plug into each other)
- Risks: high leverage, liquidity mismatches, built-in interconnectedness, no shock absorber Aramonte, Huang & Schrimpf (2021)
 - ▶ “Decentralisation illusion”: governance tends to concentrate power Aramonte, Huang & Schrimpf (2021)

Total Value Locked in DeFi

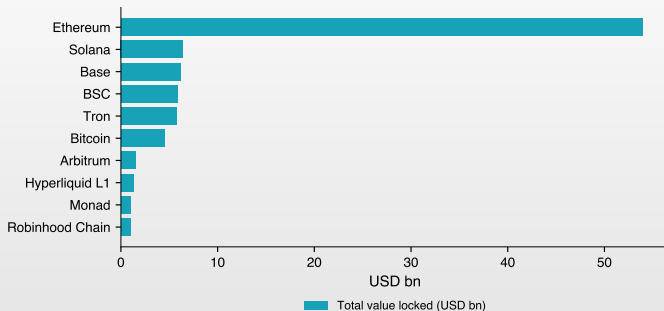


□ DeFi total value locked, all chains ([DeFiLlama](#)), and the Ethereum price, month ends, both 100 in January 2020, log scale

Reading the TVL Chart

- TVL: 0.6 billion USD in January 2020, 177 billion at the peak (9 Nov 2021), 88 billion today (50% below the peak)
 - ▶ The “DeFi summer” of 2020 multiplied TVL far faster than the Ether price
- Monthly log changes: $\Delta \ln \text{TVL}_t = a + b \Delta \ln P_t^{\text{ETH}} + u_t$ with Newey–West standard errors
 - ▶ $\hat{b} = 0.79$ (SE 0.07), $R^2 = 0.62$, $n = 80$ months
- TVL is mostly a price effect: deposits are valued at market prices, so a falling Ether mechanically lowers TVL
- A better measure of activity: TVL in units of Ether, or the fees paid by users

Where the Value Is Locked



□ Total value locked by blockchain, the ten largest, September 2026 ([DefiLlama](#))

Reading the Chain Chart

- ▣ Ethereum holds 54.0 billion USD, 56% of the 96 billion across all chains
- ▣ Next: Solana 6.4, Base 6.2, BSC (BNB Smart Chain) 5.9, Tron 5.7, Bitcoin 4.5 billion
- ▣ Base and Arbitrum are “layer-2” networks that settle on Ethereum: cheaper transactions, same security anchor
- ▣ Concentration again: one settlement layer carries most of DeFi

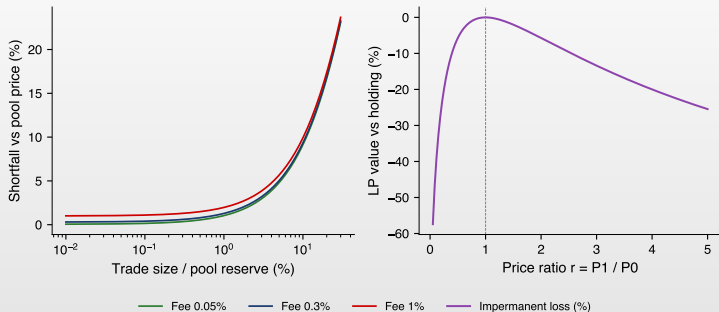
The Constant-Product Market Maker

- A pool holds reserves x of token X (e.g. Ether) and y of token Y (e.g. USD Coin); it accepts any trade that keeps $x \cdot y = k$ [Adams, Zinsmeister & Robinson \(2020\)](#)
 - ▶ Pool price (marginal): $p = y/x$ units of Y per X
- A trader sells Δx ; the pool keeps a fee f , so $\gamma \Delta x$ enters the invariant, $\gamma = 1 - f$
 - ▶ $(x + \gamma \Delta x)(y - \Delta y) = xy \Rightarrow \Delta y = \frac{y \gamma \Delta x}{x + \gamma \Delta x}$
- Execution price $\Delta y / \Delta x = p \frac{\gamma}{1 + \gamma \delta}$, with $\delta = \Delta x / x$ the trade size relative to the reserve
 - ▶ **Shortfall** against the pool price: $1 - \gamma / (1 + \gamma \delta)$: fee plus price impact
- Arbitrageurs keep p in line with other exchanges: the pool price follows the market [Angeris et al. \(2019\)](#)

Worked Example: a Swap in an Ether Pool

- Pool: $x = 2\,000$ Ether, $y = 5\,000\,000$ USD Coin, so $p = 2\,500$ and $k = 10\,000 \times 10^6$; fee $f = 0.3\%$
 - ▶ Sell $\Delta x = 20$ Ether ($\delta = 1\%$); after the fee, $\gamma \Delta x = 19.94$
- $\Delta y = \frac{5\,000\,000 \times 19.94}{2\,000 + 19.94} = 49\,358$ USD Coin, an execution price of 2467.9 per Ether
- Shortfall: $1 - 2467.9/2\,500 = 1.28\%$ (0.3% fee plus about 1% price impact)
- New pool price: $(5\,000\,000 - \Delta y)/2\,020 = 2450.8$, i.e. -1.97%
 - ▶ An arbitrageur now buys Ether in the pool and sells it elsewhere until p is back at the market price

Slippage and Impermanent Loss



- Left: shortfall against the pool price by trade size and fee; right: value of a liquidity position relative to holding the two tokens

Reading the Slippage Panel

- With a 0.3% fee: shortfall 0.40% for a trade of 0.1% of the reserve, 1.28% for 1%, 9.34% for 10%
 - ▶ Small trades pay the fee; large trades pay the price impact, roughly δ for small δ
- Lower fee tiers (0.05%) suit stable pairs; at $\delta = 1\%$ the shortfall is 1.04%
- Deep pools (large x) mean small δ : liquidity is measured by the reserve, not by an order book (Chapter 10)
- Concentrated liquidity (Uniswap v3) lets providers place reserves in a price range, deepening the pool near the current price (v3 white paper)

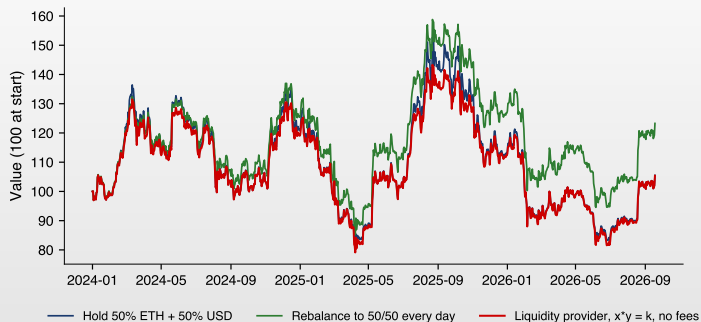
Impermanent Loss, Step by Step

- A **liquidity provider** (LP) deposits x_0 and $y_0 = p_0 x_0$ (half and half in value) and receives a share of the pool
 - ▶ Arbitrage moves the pool to the market price p_1 : from $xy = k$ and $y/x = p_1$, $x_1 = \sqrt{k/p_1}$, $y_1 = \sqrt{k p_1}$
- LP value: $V_{LP} = p_1 x_1 + y_1 = 2\sqrt{k p_1}$; holding the tokens: $V_H = p_1 x_0 + y_0$
- With $r = p_1/p_0$ and $k = p_0 x_0^2$: **impermanent loss** $IL(r) = \frac{V_{LP}}{V_H} - 1 = \frac{2\sqrt{r}}{1+r} - 1 \leq 0$
 - ▶ $r = 2$ or $r = 0.5$: -5.72%; $r = 4$ or 0.25 : -20.00%; symmetric in $\ln r$
- “Impermanent”: it vanishes if the price returns to p_0 ; it becomes permanent when the LP withdraws

Loss Versus Rebalancing

- A fairer benchmark than holding: a portfolio **rebalanced** to the pool's 50/50 mix at market prices [Milionis et al. \(2022\)](#)
 - ▶ The pool rebalances too, but at stale prices set by arbitrageurs: it always trades at a loss to them
- For $x \cdot y = k$ and a price with volatility σ , the loss versus rebalancing (LVR) grows at the rate
 - ▶ $\frac{\text{LVR}}{V} = \frac{\sigma^2}{8}$ per unit of time [Milionis et al. \(2022\)](#)
- Fees must exceed $\sigma^2/8$ a year for a liquidity provider to earn anything: a volatility-selling business
- Next chart: an Ether/USD pool since January 2024, without fees

A Liquidity Provider in an Ether Pool



- Daily Ether prices from 1 January 2024; 100 invested half in Ether, half in USD; pool with no fees

Reading the Liquidity-Provider Chart

- Ether: 2352 USD at the start, 2611 at the end ($r = 1.11$): the impermanent loss is only -0.14%
 - ▶ Final values: holding 105.5, liquidity provider 105.4, daily rebalancing 123.3
- Against rebalancing the provider lost 5.79% a year; theory: $\sigma^2/8 = 0.463/8 = 5.79\%$ with $\sigma = 68.1\%$
 - ▶ The formula matches the data almost exactly
- Impermanent loss is small when the price returns; LVR accumulates with every price move
- Break-even fee income: about 5.79% of the position a year in this period

Recap: DeFi and AMMs

- DeFi is a stack of smart contracts on a settlement blockchain; its risks are leverage, interconnection and governance
- TVL follows the Ether price: read it in units of Ether, not only in USD
- Constant product $xy = k$: execution price $p\gamma/(1 + \gamma\delta)$; impermanent loss $2\sqrt{r}/(1 + r) - 1$
- Providing liquidity is selling volatility: fees must beat $\sigma^2/8$ a year

From Bitcoin ATMs to Exchange-Traded Funds



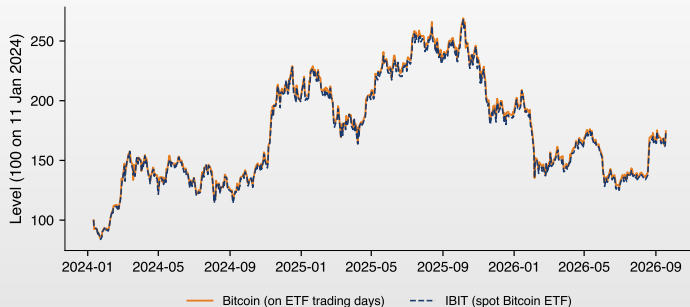
A Bitcoin ATM in Zadar, Croatia (2023)

Photo: Mnalis (2023); CC BY 4.0; Wikimedia

[Commons](#)

- Ways to hold crypto: a private wallet, a crypto exchange, a Bitcoin ATM, or a fund in a brokerage account
- A **spot ETF** holds the coins with a custodian; shares trade on a stock exchange during US hours
 - ▶ **Authorised participants** (AP) create or redeem shares in large baskets, which keeps the price close to the net asset value (NAV)
- BlackRock's IBIT starts in our data on 11 Jan 2024; sponsor fee 0.25% a year [BlackRock \(2026\)](#)
- Same arbitrage logic as a stablecoin: a band set by creation and redemption costs (Seminar 16, A8)

IBIT Follows Bitcoin

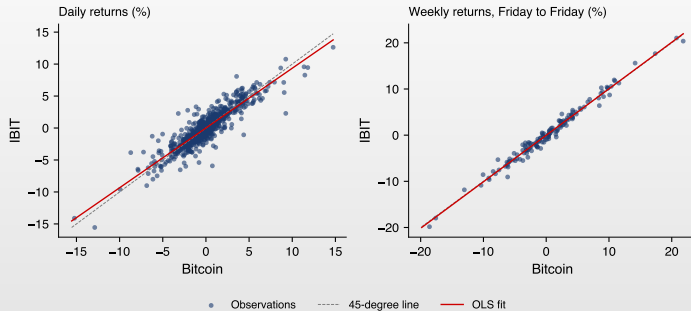


□ Bitcoin and IBIT on ETF trading days, both 100 on 11 Jan 2024

Measuring How Well an ETF Tracks

- Regression of ETF on coin returns, on the ETF's trading days: $r_t^{\text{ETF}} = a + \beta r_t^{\text{coin}} + u_t$
 - ▶ A perfect tracker: $\beta = 1$, $R^2 = 1$
- **Tracking error:** $\text{TE} = \text{sd}(r_t^{\text{ETF}} - r_t^{\text{coin}})\sqrt{m}$, with $m = 252$ (daily) or 52 (weekly)
- **Fee drift:** slope of $\ln(P_t^{\text{ETF}}/P_t^{\text{coin}})$ on time in years
 - ▶ The fund sells a little Bitcoin every day to pay its fee: the coins per share shrink at the fee rate
- **Timing:** the coin's daily close and the ETF's close are recorded at different times of the day

Daily vs Weekly Returns



- IBIT against Bitcoin: daily returns on common days (left) and Friday-to-Friday returns (right)

Reading the Tracking Results

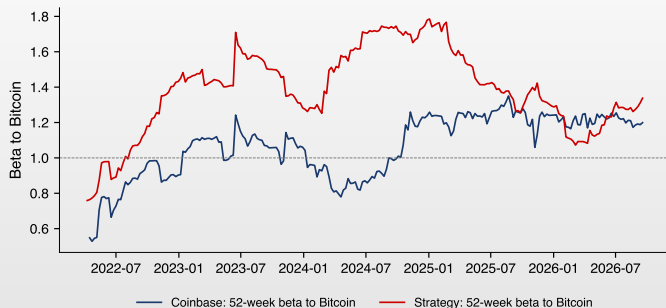
	$\hat{\beta}$ daily	R^2 daily	$\hat{\beta}$ weekly	R^2 weekly	TE daily	TE weekly
IBIT / Bitcoin	0.93	0.82	1.01	0.98	20.8	6.2
ETHA / Ether	0.94	0.83	0.99	0.98	29.9	9.0

- Daily data understate tracking ($R^2 = 0.82$, TE 20.8%); weekly data show a near-perfect tracker ($R^2 = 0.98$, TE 6.2%)
 - ▶ The daily gap is a timing mismatch between two closing prices, which averages out over a week
- Fee drift: -0.24% a year for IBIT and -0.24% for ETHA, close to the 0.25% fee
- IBIT trades 2.2 billion USD a day on average (maximum 10.3 billion on 5 Feb 2026)

Crypto Equities: Coinbase and Strategy

- **Coinbase**: a US crypto exchange; revenue depends on trading volume, which rises with prices
 - ▶ **Strategy** (formerly MicroStrategy): a software company that holds Bitcoin as its main treasury asset
- Weekly two-factor regression with Newey–West standard errors, April 2021 – September 2026
 - ▶ $r_t^{\text{stock}} = a + \beta_{\text{BTC}} r_t^{\text{BTC}} + \beta_{\text{SPX}} r_t^{\text{S\&P}} + u_t$
- Weekly returns avoid the timing mismatch between the 7-day crypto market and the 5-day stock market
- A stock with $\beta_{\text{BTC}} > 1$ is a leveraged Bitcoin position with extra equity risk

Beta to Bitcoin Over Time



□ Rolling 52-week beta of weekly returns on Bitcoin weekly returns

Reading the Crypto-Equity Results

	$\hat{\beta}_{\text{BTC}}$	SE	$\hat{\beta}_{\text{SPX}}$	SE	R^2	σ_a (%)
Coinbase	0.79	0.09	2.05	0.22	0.52	84
Strategy	1.18	0.11	1.02	0.24	0.63	89

- Strategy moves 1.18 times Bitcoin plus 1.02 times the S&P 500; its rolling beta ranged from 0.76 to 1.79
- Coinbase is more an equity than a Bitcoin proxy: $\hat{\beta}_{\text{SPX}} = 2.05$, a high-beta US stock whose business depends on crypto activity
- Both are more volatile than Bitcoin itself (55% a year on the same weeks)

Recap: ETFs and Crypto Equities

- ▣ Spot ETFs track the coin almost perfectly once timing is handled (weekly R^2 about 0.98)
- ▣ The cost is a fee drift of about 0.25% a year, visible directly in the price ratio
- ▣ Daily correlations between assets traded at different hours are biased downwards: use weekly returns
- ▣ Crypto equities mix Bitcoin exposure with equity-market risk and company-specific leverage

What Drives Crypto Prices?

- Liu & Tsyvinski (2021): crypto returns have little exposure to stock, currency or commodity factors; they respond to crypto momentum and investor attention
 - ▶ Liu, Tsyvinski & Wu (2022): three crypto factors (market, size, momentum) explain the cross-section of coin returns
- Network value: Pele & Mazurencu-Marinescu-Pele (2019a) find that Metcalfe's law (value proportional to the square of the number of users) holds in the long run but not in the short run
- Token economics: platform adoption and token value reinforce each other Cong, Li & Wang (2021)
- Arbitrage across exchanges is imperfect: large, recurrent price gaps, much larger across countries than within them Makarov & Schoar (2020)

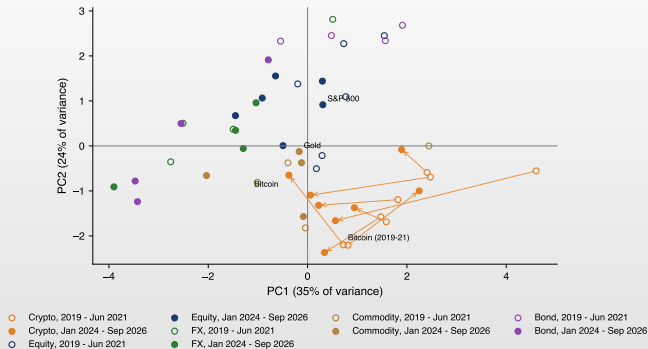
A Separate Asset Class? The Evidence So Far

- [Pele et al. \(2023\)](#) compare cryptos with stocks, exchange rates, commodities, bonds and real-estate indices
 - ▶ Dimension reduction: most of the variation is explained by **tail**, **memory** and **moment** factors of log returns
 - ▶ Classification methods place cryptos in a separate class, mainly because of the tail factor
 - ▶ Over time cryptos became more alike and more distinct from other assets
- Hedge or safe haven? A hedge is uncorrelated or negatively correlated on average; a safe haven on the worst market days [Baur & Lucey \(2010\)](#)
- Early evidence: Bitcoin a diversifier, a hedge only in some periods [Bouri et al. \(2017\)](#); we test it with post-ETF data in Seminar 16, Part C

A Small Replication: Features and Method

- 26 assets: 8 cryptos, 6 equity indices (including BET), 4 exchange rates, 4 commodities, 4 bond ETFs; two windows
 - ▶ W1: January 2019 – June 2021; W2: 11 January 2024 – 18 September 2026 (after the spot ETFs)
- Eight features of daily log returns, each asset on its own calendar
 - ▶ Moments: $\ln \sigma_a$, skewness, $\ln(1 + K)$; memory: $\hat{\rho}_1(r)$, $\hat{\rho}_1(r^2)$
 - ▶ Tails: Hill indices of losses and gains (5% largest); maximum drawdown
- **Principal component analysis (PCA)** of the standardised features (Chapter 3): directions of maximum variance
 - ▶ Distance between class centroids in the full 8-dimensional standardised space

Cryptos Among Other Asset Classes



- First two principal components; hollow markers: 2019 – June 2021, filled: January 2024 – September 2026; arrows: each crypto from W1 to W2

Reading the Asset-Class Map

- PC1 and PC2 explain 35% and 24% of the variance; PC1 mainly measures volatility, kurtosis, drawdown depth and tail heaviness
 - ▶ Cryptos sit apart in both windows, as in [Pele et al. \(2023\)](#)
- But they moved **towards** the other classes after 2024: distance to equities from 3.03 to 2.63, to commodities from 2.26 to 1.86
 - ▶ Bitcoin's volatility fell from 77% to 47%; median crypto kurtosis from 16.9 to 4.0
- Cryptos also became more alike: average distance to the crypto centre from 2.13 to 1.43
- Two windows and 26 assets are a small sample: the published study uses many more assets and a formal classification

Recap: Alternative Assets

- ▣ Crypto returns are driven by crypto-specific factors (market, size, momentum, attention), not by stock factors
- ▣ Tail, memory and moment features separate cryptos from other asset classes [Pele et al. \(2023\)](#)
- ▣ Since the spot ETFs cryptos are calmer and closer to equities and commodities, yet still distinct
- ▣ Positive correlation with shares since 2020: a diversifier at best, not a safe haven by default

Risk in Numbers: Crypto Next to Shares

	VaR 1% daily	ES 2.5% daily	worst day	MDD
Bitcoin	10.2%	10.3%	−46.5%	−82%
Ethereum	13.5%	13.7%	−55.1%	−94%
Solana	14.5%	17.1%	−55.0%	−96%
Dogecoin	13.8%	16.4%	−51.5%	−92%
S&P 500	3.4%	3.8%	−12.8%	−34%
Gold	3.0%	3.2%	−11.1%	−28%

- Historical, 2018–2026; VaR 1% and ES 2.5% as in Chapter 7 (losses as positive numbers)
- A 1 million EUR Bitcoin position needs a one-day ES 2.5% buffer of about 10.3%, roughly three times an S&P 500 position
- Filtered and extreme-value methods (Chapter 7) and their backtests (Chapter 8) apply unchanged; for Bitcoin, high-frequency entropy also helps forecast daily VaR [Pele & Mazurencu-Marinescu-Pele \(2019b\)](#)

Common Pitfalls

- ▣ Annualising crypto volatility with 252 days, or mixing 7-day and 5-day calendars in one regression
- ▣ Comparing daily closes recorded at different times and concluding that an ETF tracks badly
- ▣ Using total supply instead of circulating supply in a market-value index
- ▣ Reading TVL growth as adoption when it is mostly the Ether price
- ▣ Treating a stablecoin as risk-free cash: the peg is only as good as redemption and reserves
- ▣ Survivorship: studying only today's large coins ignores the many that failed [Będowska-Sójka, Wójcik & Pele \(2026\)](#)

Coming Next

- ▣ Chapter 17: bubbles and crashes: explosive-root tests and log-periodic power laws, with crypto booms as a main case
- ▣ Chapter 18: systemic risk and networks: spillovers between crypto, banks and equity markets
- ▣ Earlier chapters revisited: VaR and ES for crypto (Chapter 7), liquidity and order books on crypto exchanges (Chapter 10) [Agakishiev, Härdle & Pele \(2025\)](#)
- ▣ Further reading: [Härdle, Harvey & Reule \(2020\)](#), [Franke, Härdle & Hafner \(2019\)](#), [Schär \(2021\)](#)

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Are Cryptos becoming alternative Assets?
- ▣  Introduction to Blockchain and Cryptocurrencies
- ▣  Comparison of Cryptocurrency Indices
- ▣  Cryptocurrency as an Asset Class
- ▣  Smart Contracts
- ▣  Stabilising Onchain Stablecoin Risks
- ▣  Risk Premia in the Bitcoin Market

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_16

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